# GRD Supply Intelligence

**Forecast demand, spot stock-out risk, recommend action**

Silchar Fill Rate %
88.78%

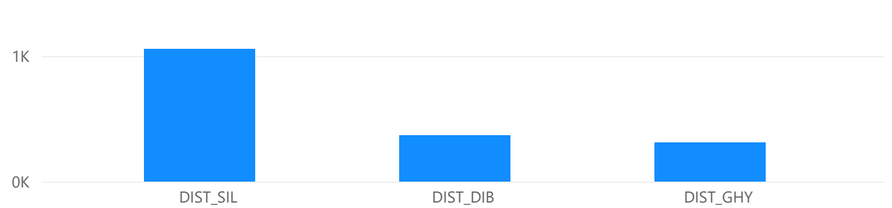
ML Improvement %
12.97%

Items Needing Action
12

Too Late Items
3

Stockout days by distributor (baseline simulation)

Stockout Days by location_id




Urgent actions, as of 5 Sept 2019

| sku_id | location_id | action | qty |
|---|---|---|---|
| M01AB | DIST_SIL | EXPEDITE warehouse shipment (express) | 10 |
| M01AE | DIST_SIL | EXPEDITE - AT RISK (no time buffer) | 10 |
| N02BE | DIST_SIL | EXPEDITE warehouse shipment (express) | 30 |

## How It Works

Forecast weekly demand → Project stock, find shortage date → Recommend order or transfer → Human approves

Silchar's 8-day monsoon lead time causes the most stockouts. ML forecast is 11.7% better than the baseline. On 5 Sept 2019 the system flagged 3 urgent items and recommended express shipments.

Demand patterns: public pharmacy dataset. Network, stock and logistics: synthetic assumptions.

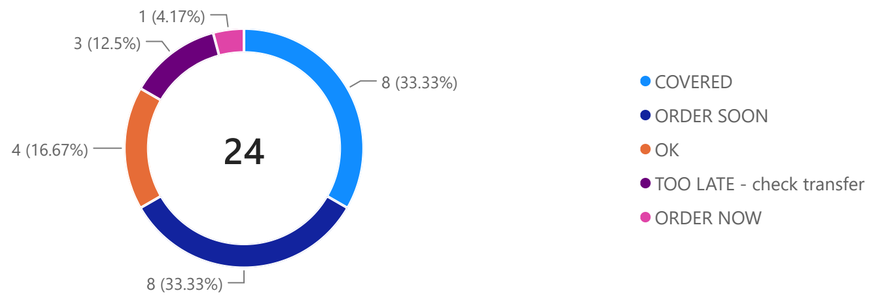
location_id
All

# Action Center

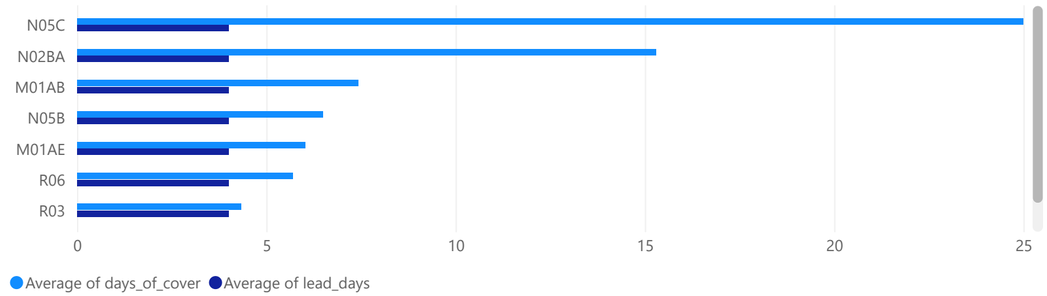
As of 5 Sept 2019, synthetic scenario

| Items Needing Action | Too Late Items | Sum of order_qty | Count of action |
|---|---|---|---|
| 12 | 3 | 133 | 3 |

**Count of sku_id**
by status




**Average of days_of_cover and Average of lead_days**
by sku_id




| location_id | COVERED | OK | ORDER NOW | ORDER SOON | TOO LATE - check transfer | **Total** |
|---|---|---|---|---|---|---|
| DIST_DIB | 3 | 1 | | 4 | | **8** |
| DIST_GHY | 5 | 1 | | 2 | | **8** |
| DIST_SIL | | 2 | 1 | 2 | 3 | **8** |
| **Total** | **8** | **4** | **1** | **8** | **3** | **24** |

| location_id | sku_id | status | days_of_cover | shortage_date | order_by | order_qty |
|---|---|---|---|---|---|---|
| DIST_SIL | M01AB | TOO LATE - check transfer | 4.80 | 11 September 2019 | 03 September 2019 | 0 |
| DIST_SIL | M01AE | TOO LATE - check transfer | 2.90 | 09 September 2019 | 01 September 2019 | 10 |
| DIST_SIL | N02BE | TOO LATE - check transfer | 4.50 | 10 September 2019 | 02 September 2019 | 30 |
| DIST_DIB | M01AB | ORDER SOON | 12.10 | 18 September 2019 | 15 September 2019 | 10 |
| DIST_DIB | N02BE | ORDER SOON | 3.50 | 16 September 2019 | 13 September 2019 | 10 |

| location_id | sku_id | action | qty | borrow_from_distributor | warehouse_express |
|---|---|---|---|---|---|
| DIST_SIL | M01AB | EXPEDITE warehouse shipment (express) | 10 | Not possible (DIST_GHY: transferable stock below one pack (need 10, spare 1, eligible 13); DIST_DIB: route not permitted) | arrives 2019-09-09, shortage 2019-09-11 |
| DIST_SIL | M01AE | EXPEDITE - AT RISK (no time buffer) | 10 | Not possible (DIST_GHY: arrives after shortage date; DIST_DIB: route not permitted) | arrives 2019-09-09, same day as shortage 2019-09-09 |
| DIST_SIL | N02BE | EXPEDITE warehouse shipment (express) | 30 | Not possible (DIST_GHY: donor has no surplus (spare -1); DIST_DIB: route not permitted) | arrives 2019-09-09, shortage 2019-09-10 |

sku_id

N02BE

# Demand Outlook

location_id

DIST_SIL

Future forecast: 4 weeks beyond the data, recursive, uncertainty grows each week

WAPE ML %

18.58%

ML Improvement %

16.20%

Forecast Bias %

1.14%

Future Forecast

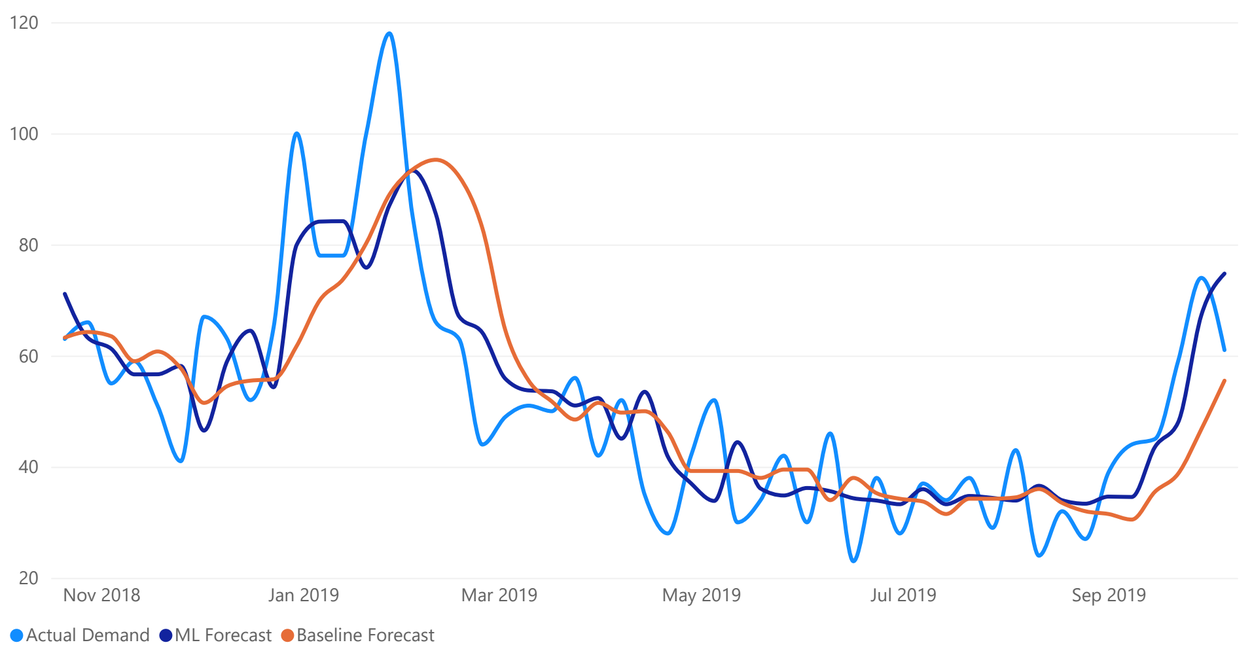
by Date




Held-out test: forecast vs actual, last 52 weeks

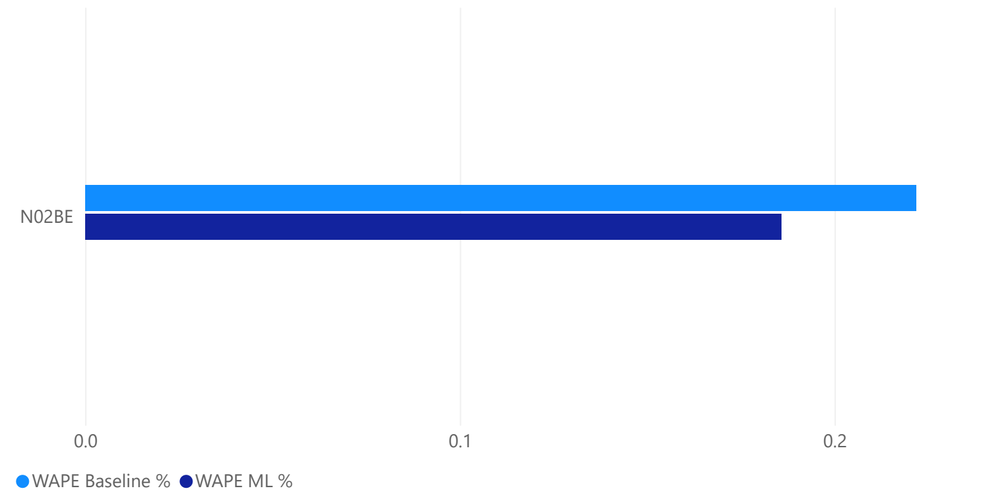
Actual Demand, ML Forecast and Baseline Forecast

by date




WAPE Baseline % and WAPE ML %

by sku_id



sku_id

All

# Performance

**Baseline simulation (before improvements): Synthetic network, assumed lead times.**

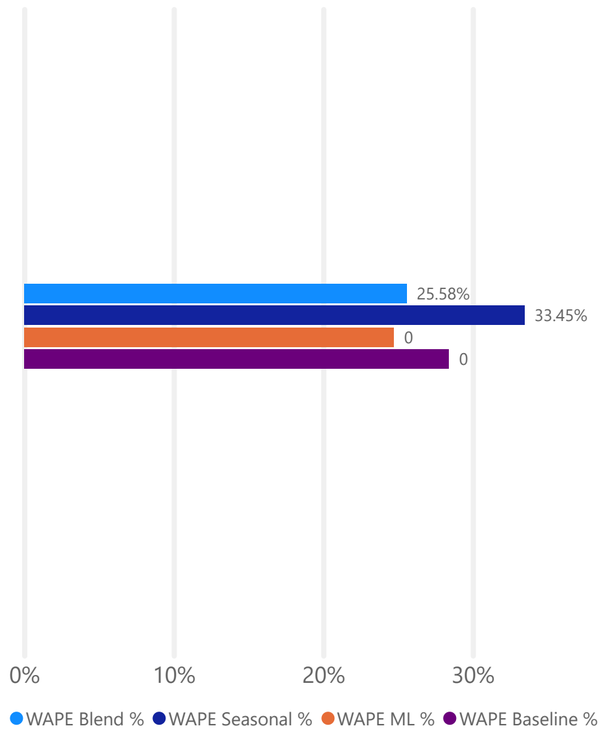
Fill Rate %
95.28%

Stockout Days (display)
1,730

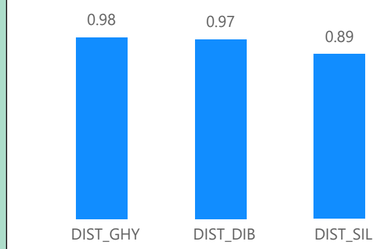
Unmet Units
6027

ML Improvement %
12.97%

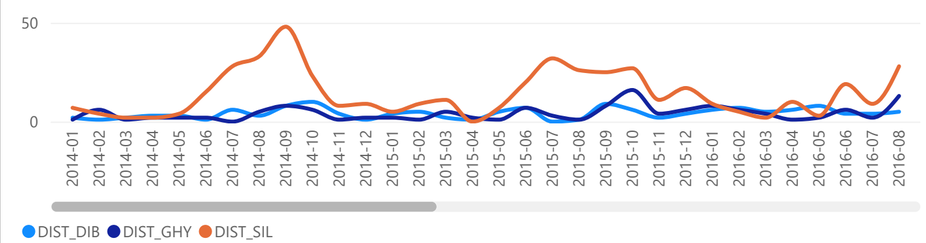
Forecast error (WAPE) by model, lower is better



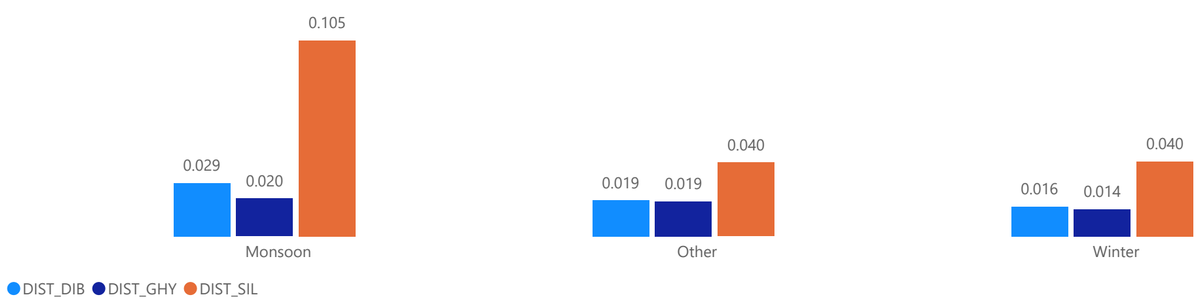

Fill Rate %
by location_id




Stockout rate by season: Silchar spikes in monsoon
Stockout Days by Year-Month and location_id




Stockout Rate
Stockout Rate % by Season and location_id




**Expired units: 0, fast-moving medicines don’t sit long enough to expire.**

# Policy Comparison

**Stockout Change vs baseline**

Stockout Change
-45.8%

**Unmet units vs baseline**

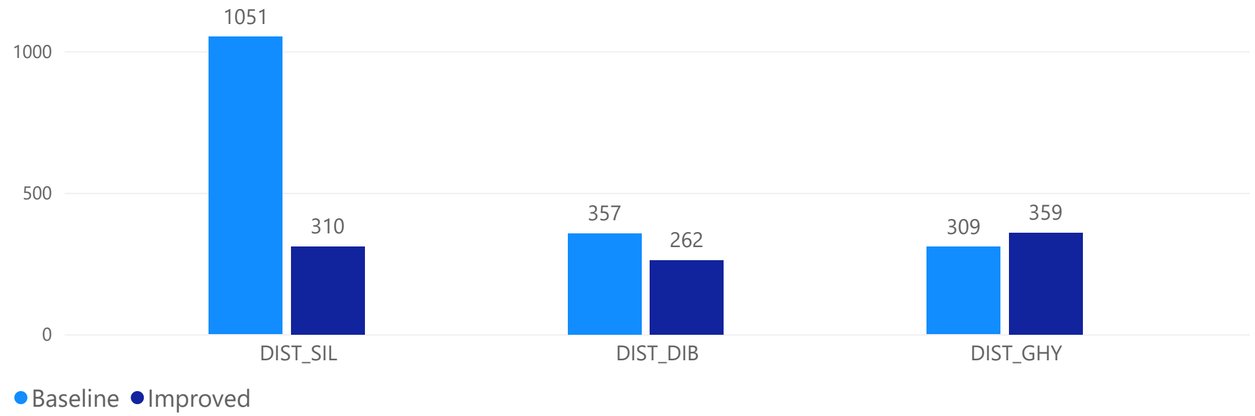
Unmet Change
-44.3%

**Inventory Change vs baseline**

Inventory Change
+24.4%

## Stockout days by distributor: baseline vs improved

Baseline and Improved by location_id




| KPI | baseline | improved | change_% |
|---|---|---|---|
| avg_inventory_value | 1,814.06 | 2,256.98 | 24.42 |
| fill_rate_% | 95.06 | 97.25 | 2.30 |
| stockout_days | 1,717.00 | 931.00 | -45.78 |
| unmet_units | 6,305.00 | 3,511.00 | -44.31 |

- Simulation on synthetic data, same demand and delivery delays for both policies
- Silchar improved most (1051 → 310); Guwahati got slightly worse (safety stock cut from 2 to 1 day)
- Trade-off: 24% more inventory
- Baseline here is re-run with shared random delivery delays, so it differs slightly from the Performance page.